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Recommendation letter for Tianlun HE

To Whom It May Concern,

It is my great pleasure to write in strong support of Mr. HE Tianlun's application to your PhD program. I have known Tianlun for three years as his academic advisor in the Integrative Systems and Design program at HKUST, where we meet every semester to discuss his progress. I have also taught him in two project-based courses and supervised his Undergraduate Research Opportunities Program (UROP) project in my lab. Across these interactions, he stands out as one of the most capable and self-driven undergraduates I have worked with.

His greatest strength is the ability to take a system all the way onto real hardware. Whether in his UROP project on a hydraulic soft robotic arm or in my courses, where he built a working 6-DOF underwater robot rather than a simulation, he consistently gets perception and control algorithms running on physical robots and solves problems independently, with a maturity unusual for a third-year undergraduate. This is exactly the profile I look for in a doctoral researcher in embodied intelligence.

Beyond technical skill, Tianlun is thoughtful, reliable, and collaborates generously with peers and PhD students. I understand he has since moved toward algorithmic research—vision-language-action models, world models, and reinforcement learning on real robots—which fits him well: he already knows how to make things work on physical hardware, and few researchers combine that with a genuine interest in the underlying algorithms. A PhD is the natural next step, and I recommend him with my highest confidence. Please feel free to contact me if any further information would be helpful.

Sincerely yours,

Prof. Rob Scharff,
Assistant Professor
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